

1. OVERVIEW

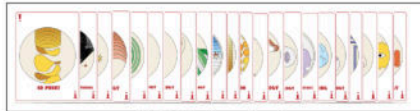
Heritage Spark



You can use the cards to facilitate discussion by immersing participants in the characters, understanding the problem space, and generating ideas.

The cards provide you with guidelines to facilitate digital designs for ICH across the three categories shown below:

2. Contents



23 Technology Cards



10 ICH Cards



7 Character Cards



2 Thinking Canvases

3. INSTRUCTIONS

How are the cards organized?

Each category is divided into three types of cards

Technology Cards

- Description
- case
- Advantages&disadvantages

Character Cards

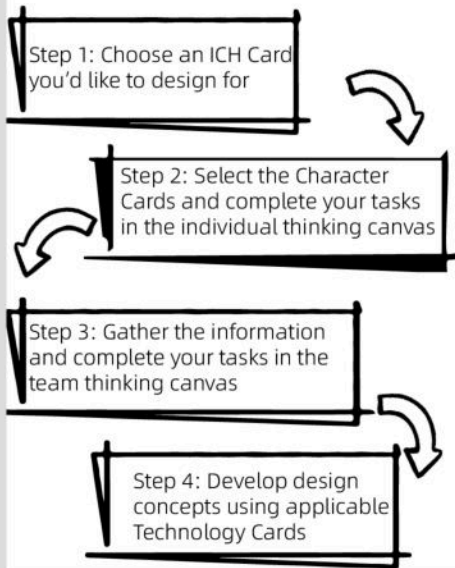
- character introduction
- goals
- stage tasks

ICH Cards

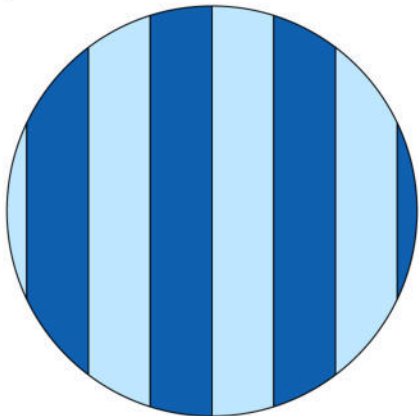
- Description
- detail
- case



4. How to use the cards?



C.



民间文学

Folk Literature

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Overview

Folk literature includes folktales, legends, proverbs, ballads, etc. From ancient myths and legends to modern folk tales, folk literature widely circulates among people with vivid narrations and profound educational implications, enriching people's cultural life.

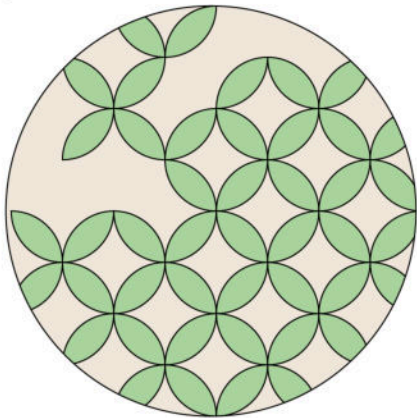
Artistic Features

Its language is vivid, plots are complicated, characters are lifelike and meaningful, and these works often carry strong local and ethnic characteristics.

Great Works

Such as the mythical stories in the "Shan Hai Jing", short stories in "Liaozhai Zhiyi" and various local folk legends and historical novels.

C.



民俗

Folk Customs

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Overview

Folk customs include traditional festivals, rituals, folktales, and beliefs. With a long history in China, they originated from ancient production activities and religious beliefs and are a direct reflection of societal culture.

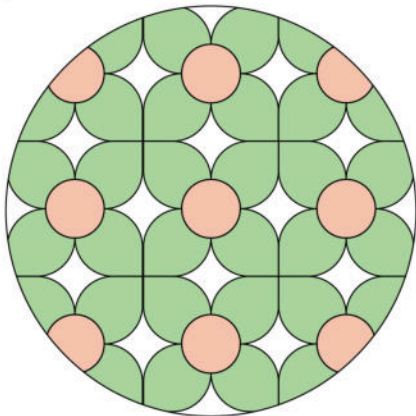
Artistic Features

Folk customs is rich and diverse, including the dragon dance and lion dance during the Spring Festival, as well as moon gazing during the Mid-Autumn Festival.

Great Works

These include dragon boat races during the Dragon Boat Festival and tomb sweeping during the Qingming Festival.

C.



传统医药

Chinese Medicine

i



Overview

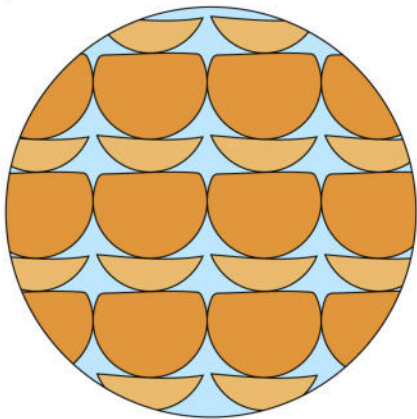
Traditional medicine includes traditional Chinese medicine and various traditional treatment methods such as acupuncture, cupping, etc. It is based on classical medical texts such as "Huangdi Neijing" and is an important component of ancient Chinese science.

Artistic Features

It emphasizes the balance of yin and yang and focuses on holistic treatment of the body and mind.

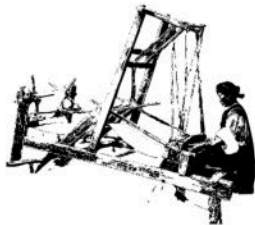
Great Works

Such as traditional Chinese medicine formulas and acupuncture treatment methods.



传统技艺

Traditional Craftsmanship



Overview

Traditional craftsmanship includes handicraft techniques like ceramic making, silk manufacturing, wood carving, stone carving, and more. This craftsmanship has been an important part of Chinese culture since ancient times, passed down through generations with meticulous care.

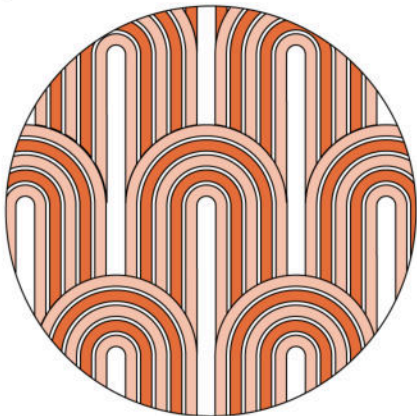
Artistic Features

Traditional crafts emphasize the selection and treatment of materials, the precision and innovation of techniques, and the design of products.

Great Works

Examples of great works include Jingdezhen porcelain, Su embroidery, and Hangzhou silk weaving.

C.



传统美术

Traditional Fine Art

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overview

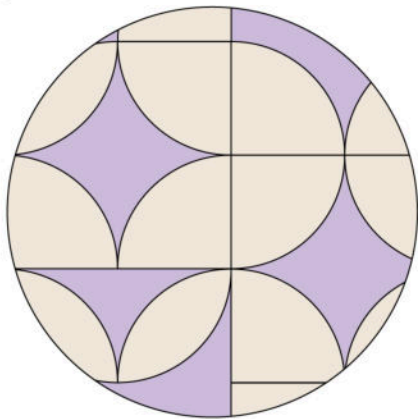
Traditional fine arts include various forms such as calligraphy, painting, sculpture, seal carving, and more, and are an important component of Chinese culture. From ancient wall paintings to calligraphic seal carvings, each art form has its unique heritage and lineage.

Artistic Features

Traditional arts emphasize the beauty of lines, harmony of colors, and profound artistic conception.

Great Works

Examples of great works include Wang Xizhi's "The Orchid Pavilion" and Zhang Dayan's ink-wash paintings.



传统体育、游艺与杂技

Traditional Sports, Games and Acrobatics



Overview

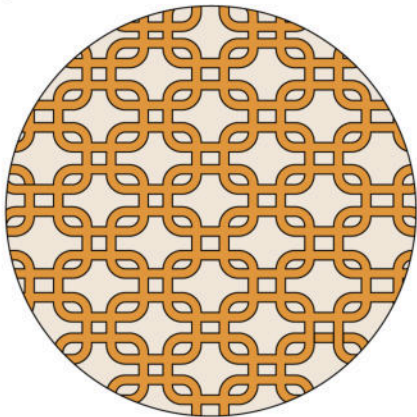
Traditional Chinese sports, games, and acrobatic performances include activities such as Tai Chi, Chinese chess, and various acrobatic acts. These activities have a deep historical foundation and were integral to the pastoral and entertainment life of ancient people.

Artistic Features

Traditional sports emphasize the harmony of body and spirit, amusements combine wisdom and strategy, and acrobatics showcase human limits and artistic beauty.

Great Works

Notable examples include Tai Chi performances, Chinese chess, and acrobatic programs like balancing acts and needle-walking.



曲艺
Quyi



Overview

Quyi includes many forms, including rap, opera excerpts, and stand-up comedy. Originating from ancient folk storytelling, Quyi has a history spanning thousands of years.

Artistic Features

Quyi primarily employs ventriloquism, singing, recitation, and acting as its main performance forms. It features a variety of styles and is rich in local characteristics and ethnic customs.

Great Works

Some of the great works of Quyi include crosstalk, storytelling, Beijing Opera Drum, and Suzhoupingtan.



传统戏剧

Traditional Theater



Overview

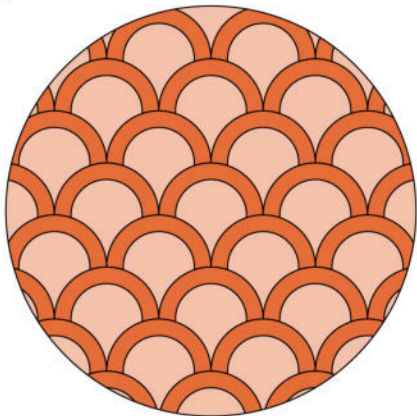
Traditional theater includes forms like Peking Opera, Yue Opera, and Cantonese Opera, which have developed from folk performances and embody Chinese dramatic culture.

Artistic Features

Traditional drama combines singing, reciting, acting, and fighting, and has unique masks, costumes and performance techniques.

Great Works

Examples include "The Injustice of Dou E" and "A Dream of Red Mansions."



传统舞蹈

Traditional Dance



Overview

Traditional dance includes ethnic, folk, and classical dances. With a rich tradition in Chinese history, dance has evolved from ancient religious rituals to modern stage performances, constantly changing in style and form.

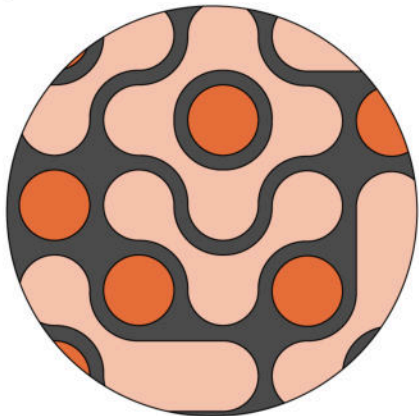
Artistic Features

Each dance has its own unique costumes, movements and accompaniment, such as Mongolian dance, and Tibetan dance.

Great Works

Examples include dance performances in Peking Opera such as "The Drunken Concubine" and traditional dances of ethnic minorities like the Fire Torch Festival dance of the Yi people in Yunnan.

C.



传统音乐

Traditional Music

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Overview

Traditional music includes classical music, folk music, and various ethnic music forms. From ancient court music to various folk songs and storytelling, each style has its unique development history.

Artistic Features

Traditional music is known for its unique instruments, music structures, and performance forms, such as the use of instruments like erhu, pipa, and dizi.

Great Works

Such as classic pieces like "Mountain Melodies and Rushing Water," and "Embattled Conflict in Every Direction."



媒体

Media

Character Introduction:

The media is responsible for driving information dissemination and shaping public opinion. Your role involves collecting, organizing, and publishing information on intangible cultural heritage, expanding its influence, and stimulating interest and discussion across society.

Target:

Media should support heritage inheritors through research and enhance public understanding via reporting and content production. This includes providing public opinion support and promoting modern transformation.

Task 1: Research

- **Attitude Survey:** Gauge public views, awareness, and expectations of intangible cultural heritage, and assess its impact on local and broader society, including economic, educational, and cultural aspects.
- **Channel Research:** Explore various communication channels and content formats to identify the most effective methods for enhancing the visibility and appeal of intangible cultural heritage projects.

Tip: Incorporate the Media's perspective and emotions to resonate with the audience.

Task 2: Problem Definition

Stakeholders' Meeting: Exchange research information with other stakeholders and summarize the pain points of intangible cultural heritage projects.

Task 3: Brainstorming

Brainstorming: Use the technology cards to discover innovative solutions for intangible cultural heritage and write down your ideas.



企业
Enterprise

Character Introduction:

Enterprises play a key role in innovating intangible cultural heritage by applying resources, market expertise, and innovative thinking to explore commercial potential and promote sustainable development.

Target:

Enterprises should support heritage inheritors through research and focus on developing products or services that respect tradition while adapting to market demands through innovative business models.

Task 1: Research

- **Market Analysis:** Assess the project's market performance and potential customer base. Study existing related products or services and explore opportunities for improvement and development.
- **Model Exploration:** Analyze how various business models can enhance commercial value and facilitate the transformation of intangible cultural heritage.

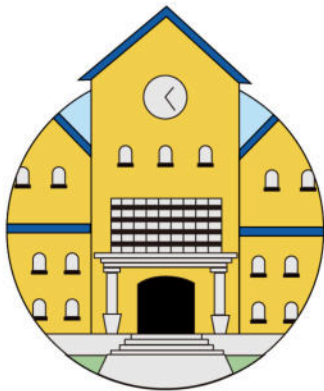
Tip: Incorporate the enterprises' perspectives and emotions to resonate with problem definitions.

Task 2: Problem Definition

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Task 3: Brainstorming

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教育机构

Educational Institution

Character Introduction:

Educational institutions are responsible for preserving and teaching intangible cultural heritage. By integrating ancient knowledge with modern education, they play a crucial role in public education and talent cultivation.

Target:

Educational institutions are responsible for preserving and teaching intangible cultural heritage. By integrating ancient knowledge with modern education, they play a crucial role in public education and talent cultivation.

Task 1: Research

- **Current Situation Analysis:** Assess the survival status of intangible cultural heritage, including its local influence and its transmission among the younger generation.
- **Form Investigation:** Examine the existing methods of popularizing intangible cultural heritage education to understand their strengths and weaknesses.

Tip: Incorporate the perspectives and emotions of educational institutions to create resonance and engagement.

Task 2: Problem Definition

Stakeholders' Meeting: Exchange research information with other stakeholders and summarize the pain points of intangible cultural heritage projects.

Task 3: Brainstorming

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政府

Government

Character Introduction:

Government act as cultural guardians and development planners, responsible for promoting the protection of intangible cultural heritage and its integration into modern society through policymaking and resource allocation.

Target:

The government should support heritage inheritors through research, utilize its influence and resources for effective inheritance and innovation, and enhance cultural impact.

Task 1: Research

- **Policy Evaluation:** Research and evaluate existing intangible cultural heritage protection policies to understand their effectiveness and implementation shortcomings.
- **Resource Survey:** Identify government support resources, such as funds, technology, and talent for protecting and developing intangible cultural heritage.

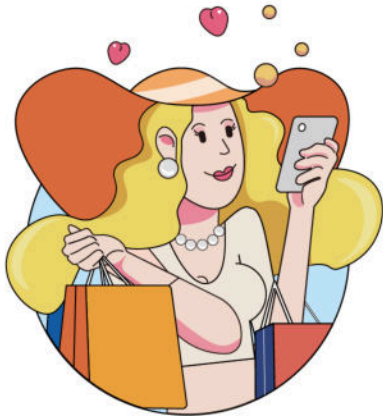
Tip: Incorporate the government's perspective and emotions to resonate with them.

Task 2: Problem Definition

Stakeholders' Meeting: Summarize information gathered from other stakeholders and organize meetings to identify and consolidate the pain points of intangible cultural heritage projects.

Task 3: Brainstorming

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消费者

Consumer

Character Introduction:

Consumers experience and evaluate intangible cultural heritage products. Their preferences and decisions influence the circulation and success of these products, with their purchasing power reflecting recognition and supporting sustainable development.

Target:

Consumers should assist heritage inheritors in researching projects and provide insights into innovative design transformations as market participants.

Task 1: Research

- **Market insights:** Analyze target consumers' attitudes, demands, and behaviors to identify trends and align product design with market needs.
- **Experience feedback:** Engage with products or services related to intangible cultural heritage and offer suggestions for improvements or identify new consumer needs.

Tip: Incorporate consumers' perspectives and emotions to resonate with them.

Task 2: Problem Definition

Stakeholders' Meeting: Summarize information gathered from other stakeholders and organize meetings to identify and consolidate the pain points of intangible cultural heritage projects.

Task 3: Brainstorming

Brainstorming: Use the technology cards to discover innovative solutions for intangible cultural heritage and write down your ideas.



技术专家

Technical Expert

Character Introduction:

Technical experts pioneer the integration of intangible cultural heritage with modern technology, utilizing design and expertise to solve problems, innovate methods of heritage transmission, and broaden the scope of heritage projects.

Target:

Technical experts must assist heritage inheritors in researching projects to ensure that the final design respects traditional culture and meets contemporary development needs.

Task 1: Research

- **Cultural Exploration:** Conduct an in-depth study of the historical and cultural value of intangible cultural heritage items to understand their significance and role in local society.
- **Technology Exploration:** Investigate and evaluate the application of technologies to protect, record, transmit, and promote intangible cultural heritage.

Tip: Incorporate the technical experts' perspectives and emotions to resonate with them.

Task 2: Problem Definition

Stakeholders' Meeting: Summarize information gathered from other stakeholders and organize meetings to identify and consolidate the pain points of intangible cultural heritage projects.

Task 3: Brainstorming

Brainstorming: Use the technology cards to discover innovative solutions for intangible cultural heritage and write down your ideas.



非遗传承人

Intangible Cultural Heritage Inheritor

Character Introduction:

Intangible cultural heritage inheritors are skilled guardians of tradition but face the challenges of declining awareness and transmission. You aim to collaborate with others to promote innovative, modern transformations of ICH.

Target:

ICH inheritors must lead a team to conduct in-depth research on a designated ICH project, ultimately proposing innovative applications to enhance the modern relevance and appeal of the cultural heritage.

Task 1: Research

- **Cultural Exploration:** Identify key components of ICH items, including forms of expression, materials, techniques, and performance methods.
- **Challenge Analysis:** Examine the current development of ICH projects, including the number of inheritors, audience scope, main challenges, and application prospects.

Tip: Incorporate the inheritors' perspectives and emotions to create a resonant connection.

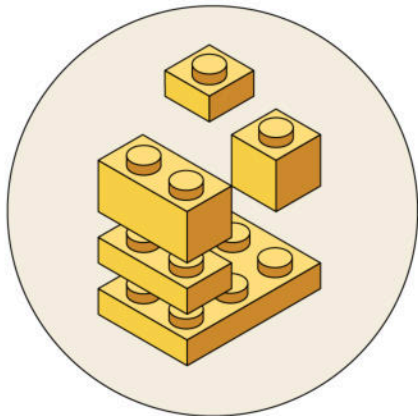
Task 2: Problem Definition

Stakeholders' Meeting: Summarize information gathered from other stakeholders and organize meetings to identify and consolidate the pain points of intangible cultural heritage projects.

Task 3: Brainstorming

Brainstorming: Use the technology cards to discover innovative solutions for intangible cultural heritage and write down your ideas.

T



MODULAR TECHNOLOGY

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Modular Technology

Modular technology involves breaking down a system or structure into standardized, independent units (modules) with specific functions. This approach allows for easier design, manufacturing, testing, maintenance, and upgrades while reducing costs.

Tips

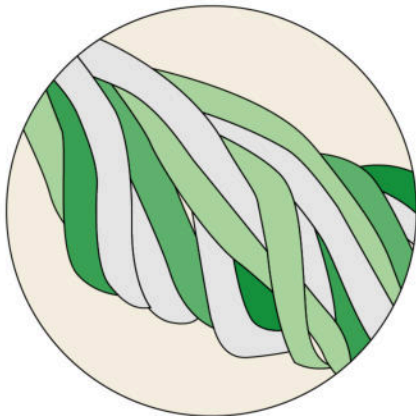
- ✓ **Flexibility and Scalability:** Enables independent design and manufacturing of components, enhancing flexibility and allowing easy adjustments or upgrades.
- ✓ **Simplified Manufacturing and Maintenance:** Standardized, interchangeable components streamline production processes, reducing manufacturing and maintenance costs.
- ✗ **Creativity Limitations:** May restrict certain creative expressions.

Case



This is a set of DIY connectors that can be assembled to create various models and objects. Each component connects directly to others, allowing for more creative variations. Building modular structures helps users effectively understand and develop three-dimensional creative insights.

T!



NEW MATERIAL TECHNOLOGY

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New Material Technology

New material technology involves developing and using advanced materials to enhance product performance and manufacturing. These materials often possess superior properties such as increased strength, improved thermal stability, reduced weight, and enhanced sustainability.

Tips

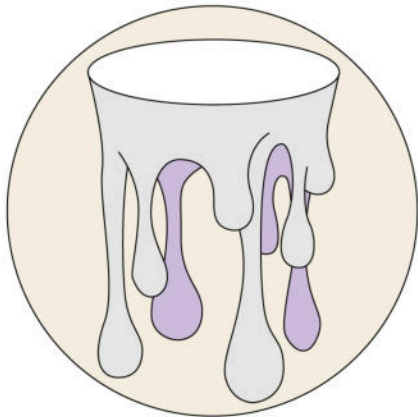
- ✓ **Enhancing Product Performance:** Push the boundaries of traditional materials by introducing novel physical and chemical properties, thereby improving innovation, durability, and longevity.
- ✓ **Environmental Impact and Sustainability:** Employing eco-friendly materials promotes environmental sustainability.
- ✗ **Safety and Health Impact:** The long-term health and environmental effects of new materials are not fully understood; comprehensive safety testing and evaluation are essential.

Case



This computer-aided design system integrates liquid crystal display materials to merge embroidery with electronic textiles. Conductive threads function as resistive heaters, activating thermochromic dyes or pigments when energized.

T!



MOLDING TECHNOLOGY

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Molding Technology

Molding technology involves heating, pressurizing, or injecting raw materials into a mold to solidify or cure them into a specific shape, ultimately forming the desired parts or products. Common molding methods include injection molding, compression molding, and blow molding.

Tips

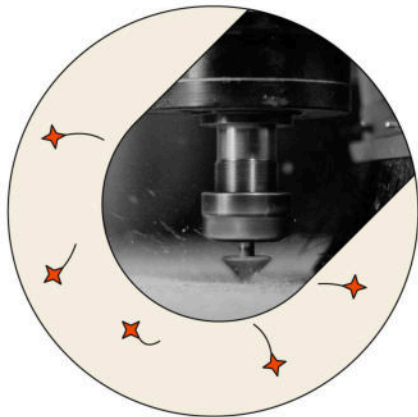
- ✓ **Production Capability:** Supports high-volume production, significantly increasing efficiency and reducing unit costs.
- ✓ **Complex Design Implementation:** Handles intricate designs, allowing for fine details and complex structures, thereby expanding design possibilities.
- ✗ **Creativity and Craft Limitations:** While molding technology enables mass replication, it may limit product uniqueness and personalization, potentially diminishing the value of traditional craftsmanship.

Case



Designers combined traditional dough modeling with automated molding, revitalizing dough flower carving. This integration ensured quality and consistency through industrial production, enhancing processing efficiency and capacity.

T



SUBTRACTIVE MANUFACTURING

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Subtractive Manufacturing

Subtractive manufacturing creates objects by removing material layer by layer using methods such as CNC machining, laser cutting, and EDM. It is commonly used for rapid prototyping, custom manufacturing, and producing functional parts.

Tips

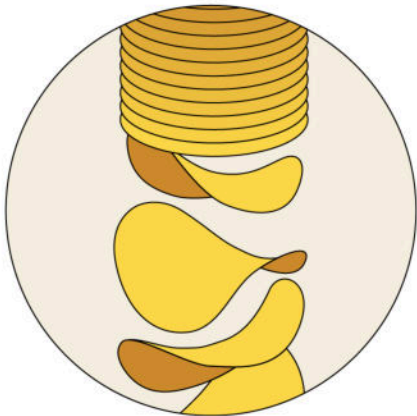
- ✓ **High Precision and Complexity:** Enables the creation of highly detailed and intricate designs, providing significant technical support for traditional intangible cultural heritage
- ✓ **Rapid Prototyping:** Allows designers to quickly iterate and innovate without traditional constraints, creating unprecedented shapes and complexities.
- ✗ **Environmental Impact:** Material removal in subtractive manufacturing can produce significant waste and dust, necessitating effective environmental controls and waste management to minimize impact.

Case



This fast mechanical seal-cutting system transforms designs into physical seals by combining software and hardware to enhance efficiency and lower technical barriers. It offers new possibilities for preserving and reviving intangible cultural heritage.

T



4D PRINTING

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4D Printing

4D printing is an innovative manufacturing technology similar to 3D printing but allows printed objects to change shape or function over time. This technology combines traditional 3D printing with smart materials, enabling objects to respond to temperature, humidity, light, or other environmental factors.

Tips

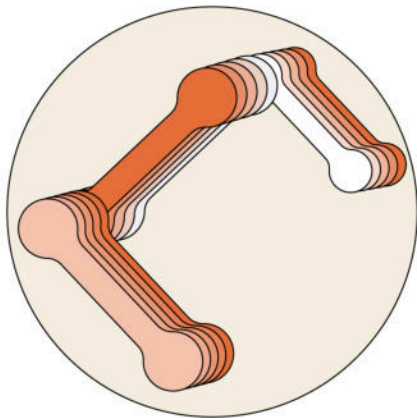
- ✓ **Innovative Design Possibilities:** Go beyond traditional manufacturing limits to explore more complex and dynamic designs.
- ✓ **Adaptive and Responsive Capabilities:** Automatically adjust shapes or functions in response to environmental changes, adding intelligence and interactivity to traditional products like clothing and home decor.
- ✗ **Material Limitations and Durability:** Printed products may not match the mechanical properties of those produced by traditional methods, especially under high-load applications.

Case



This hybrid process uses 4D printing technology and a 3D pen to achieve controllable deformation and five freehand techniques. A mixed reality system helps novices master 4D printing skills, reducing the skill threshold.

T!



ROBOTIC TECHNOLOGY

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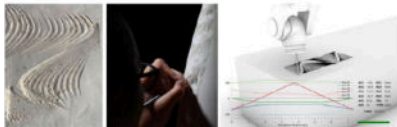
Robot Technology

Robotic technology involves the design, manufacture, operation, and control of robots. It enables automation through mechanical devices that can perform tasks or operations, often programmable, in environments where humans or other machines may not be suitable or possible.

Tips

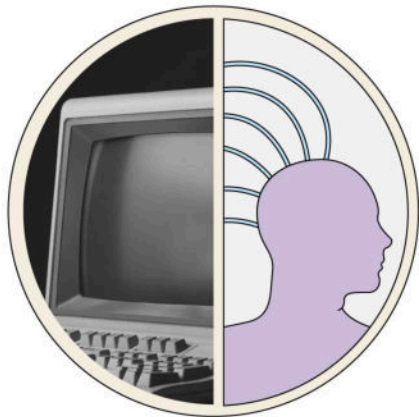
- ✓ **Improving Production Efficiency:** Robots can replace manual labor in parts or all of the production process, reducing time and cost.
- ✓ **Innovative Product Design:** Allows greater design freedom for complex creations and experiments, enabling precise execution.
- ✗ **Flexibility and Creativity Limitations:** Robots are efficient for predefined tasks but may lack human flexibility and creativity for non-standard or highly creative tasks.

Case



This industrial robot simulates the workflow of clay carving artisans. By examining the relationship between carving postures and different clay states, it demonstrates the connection between machine movements and carving details.

T



BRAIN-COMPUTER INTERFACE

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Brain-Computer Interface

Brain-computer interface (BCI) technology directly connects the human brain with external devices, such as computers or machines. It allows brain activity to control computer systems or other mechanical devices without relying on traditional input methods like keyboards and mice.

Tips

- ✓ **Enhancing Creativity:** Enable direct thought interaction with machines or systems for the immediate realization of creative ideas.
- ✗ **Accuracy and Reliability Issues:** Limitations in signal interpretation and command execution may affect precision in intangible cultural heritage production.
- ✗ **Privacy and Ethical Concerns:** Monitoring and analyzing brain activity raise significant privacy and ethical issues.

Case



The research team developed a virtual reconstruction of ancient Rome controlled by an innovative brain-computer interface system. Using non-invasive BCI technology, users can navigate a virtual character within ancient Rome and interact with other virtual characters while learning.

T!



MIXED REALITY

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Mixed Reality

Mixed Reality (MR) technology integrates elements of the real and digital worlds, creating environments where physical and virtual objects interact seamlessly. MR employs advanced visual and sensing technologies to allow real and virtual objects to coexist and interact in real-time.

Tips

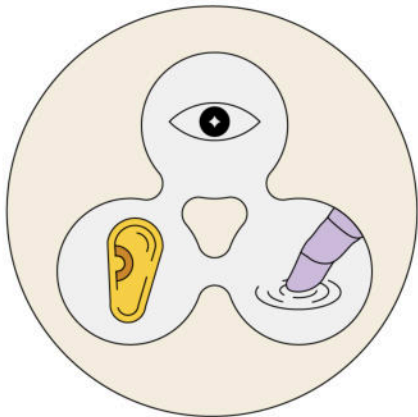
- ✓ **Enhanced Interactivity and Immersion:** Allow users to interact with virtual objects in physical space while manipulating real objects.
- ✓ **Support Real-Time Interaction and Modification:** View and adjust intangible cultural heritage designs within real environments, facilitating rapid iteration and reducing costs and time.
- ✗ **User Experience:** Current technology cannot fully merge reality and virtual worlds, affecting the completeness and authenticity of user experience.

Case



This project combines the intangible cultural heritage of Thangka with mixed technologies to create real-life Thangka displays. During panoramic tours, users can navigate intelligently using gesture and voice controls.

T!



MULTIMEDIA TECHNOLOGY

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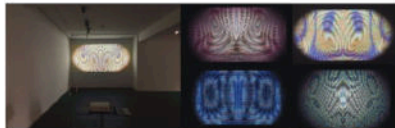
Multimedia Technology

Multimedia technology integrates text, images, sound, and video by utilizing computer systems to process, store, transmit, and display information. Its core purpose is to combine various media forms to deliver a richer and more interactive user experience.

Tips

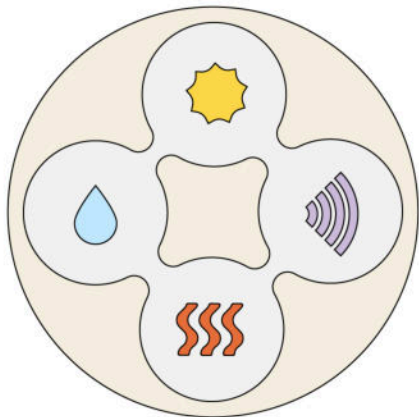
- ✓ **Enhanced Interactivity and Immersion:** Engage multiple senses—visual, auditory, and tactile—to make traditional crafts more vivid.
- ✗ **Preserving Cultural Authenticity:** Avoid excessive reliance on technology and commercialization to prevent the misinterpretation of traditional craft culture.

Case



This multimedia sound interactive installation employs algorithm-generated moiré patterns that dynamically respond to the participant's sounds, thematically reflecting the traditional Yunjin weaving process and the weavers' collaboration and effort.

T!



SENSING TECHNOLOGY

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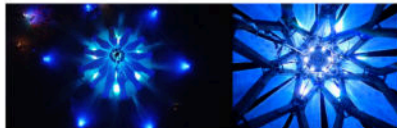
Sensing Technology

Sensor technology detects and measures physical, chemical, or biological changes, converting them into analyzable electrical signals for real-time monitoring, data collection, and automatic control.

Tips

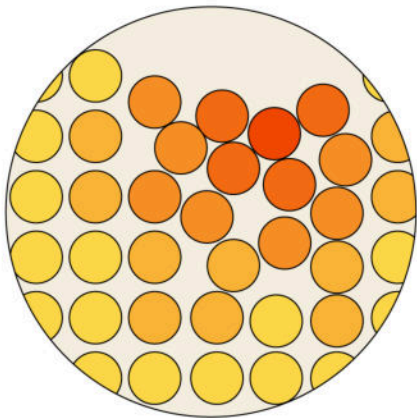
- ✓ **Precise Process Monitoring:** Sensors track key parameters such as temperature, humidity, and pressure in real-time, ensuring optimal conditions and improving product consistency.
- ✓ **Data-Driven Decision Support:** Collected data helps refine intangible cultural heritage processes, enabling more scientific and systematic innovation.
- ✗ **Data Security and Privacy:** Ensure the safety and compliance of sensitive data to prevent leaks or misuse.

Case



This modular textile sunshade combines traditional techniques with novel materials, sensors, and digital technology. It functions as a sunshade by day and transforms into an interactive art installation by night, dynamically changing colors based on environmental conditions.

T!



COMPUTER SIMULATION

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Computer Simulation

Computer simulation technology uses computers to replicate the working principles and behaviors of real systems. By building models and algorithms, it predicts system performance under various conditions and enables virtual experiments and optimization for design, testing, and training.

Tips

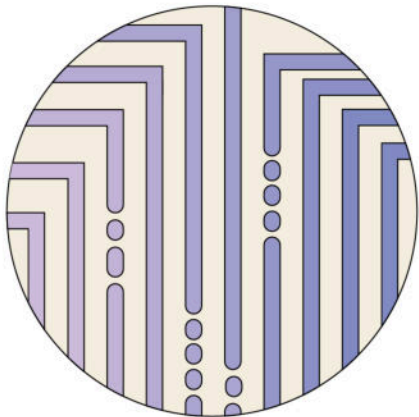
- ✓ **Accurate Prediction and Optimization:** Simulating the intangible cultural heritage process allows designers to foresee issues and optimize product parameters before production.
- ✓ **Improve Innovation and Iteration Speed:** Support rapid testing and modification of ideas without consuming a lot of materials and time.
- ✗ **Dependency and Flexibility Issues:** Over-reliance on simulation design tools may limit designers' creativity, especially in intangible cultural heritage projects where artistry and manual operations dominate.

Case



This computer-aided method generates Thangka reliefs from black-and-white line drawings by layering low, medium, and high-frequency height maps into mesh models. Designers can then edit these models to create new Thangka relief styles.

T



5G / 6G

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5G/6G

5G and 6G technologies represent the fifth and sixth generations of mobile networks, designed for higher data speeds, lower latency, and broader coverage. These technologies will drive innovation in personal devices and advance areas such as autonomous vehicles, smart cities, and telemedicine.

Tips

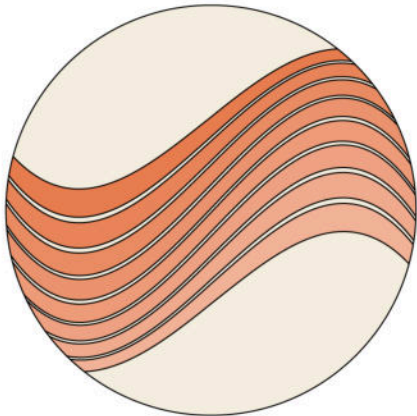
- ✓ **Enhanced Interactivity and Immersion:** 5G/6G's low latency enables real-time, immersive experiences for online intangible cultural heritage content.
- ✓ **Improved Remote Collaboration:** High bandwidth and low latency of 5G/6G support smooth, real-time collaboration between designers and heritage experts in different locations.
- ✗ **Network Coverage and Stability Issues:** Despite high speeds, 5G/6G may face coverage and signal stability problems in remote or complex environments.

Case



The Liangshan Yi Traditional Torch Festival in Xichang, China, uses 5G technology to live stream events like Carnival Night and the Yi Beauty Pageant. This approach recreates intangible cultural heritage scenes and fosters emotional resonance.

T



RF TECHNOLOGY

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RF Technology

Radio frequency (RF) technology uses electromagnetic waves for communication and transmission, including radio, radar, broadcasting, and navigation. It is employed in mobile phones, satellites, and IoT applications like Bluetooth, WiFi, NFC, and RFID.

Tips

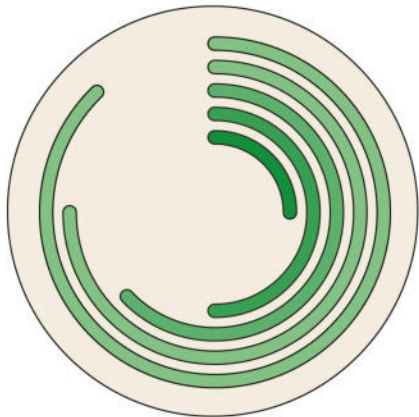
- ✓ **Enhanced Functionality and Interactivity:** Integrate antennas and RF components into traditional products to add extra features.
- ✗ **Maintenance and Reliability:** Advanced electronic components may encounter maintenance and reliability issues in harsh environments or during long-term use.

Case



FuseBot is a robotic system that leverages RF technology and introduces two key innovations: RF Vision Mapping and RF Vision Extraction. These innovations enable FuseBot to accurately locate and efficiently retrieve both RFID-tagged and untagged target objects in various occluded environments.

T



BIG DATA

i

Big Data

Big data technology includes methods and tools for processing and analyzing large, complex, rapidly changing datasets that traditional databases and software struggle to handle. Its core characteristics are the "Five Vs" : Volume, Velocity, Variety, Veracity, and Value.

Tips

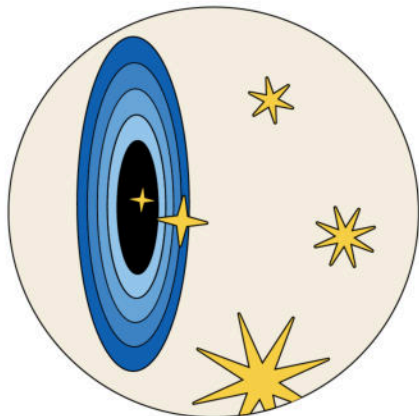
- ✓ **Market Trend Analysis:** Examines consumer behavior and trends to forecast future demand for traditional crafts.
- ✓ **Personalized Product Design:** Utilizes consumer preferences and big data to tailor craft designs, enhancing competitiveness and customer satisfaction.
- ✗ **Data Security and Privacy:** Managing sensitive data requires measures to ensure security and prevent breaches or misuse.

Case



This is an interactive visual navigation system produced by UNESCO. It leverages big data technology to provide visual navigation for nearly 500 intangible cultural heritage elements inscribed in the UNESCO 2003 Convention, visualizing the deep interconnections between them.

T



AIGC

i

AIGC

Generative artificial intelligence (AI) is a key branch of AI that learns from large datasets to mimic and create works similar to those made by humans, such as images, music, and text. This technology can not only analyze and understand data but also generate new instances based on what it has learned.

Tips

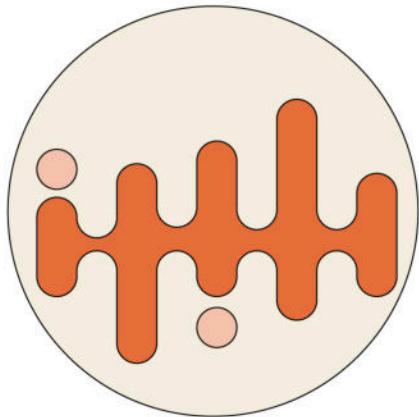
- ✓ **Innovation and Diversity:** Quickly generates numerous design options, offering innovative and diverse choices for traditional heritage products.
- ✓ **User Participation in Customization:** Allows users to personalize heritage products through AI tools, boosting satisfaction and market appeal.
- ✗ **Originality and Cultural Value:** AI-generated designs may raise concerns about originality and cultural value.

Case



PaCaNet, based on CycleGAN, merges traditional Chinese painting and calligraphy by blending visual elements from both art forms, inspired by Chinese pictograms. It offers a new approach to artistic creation.

T



VOICE TECHNOLOGY

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Voice Technology

Speech recognition technology allows computers to analyze, recognize, and synthesize sound. It involves converting human speech into text or commands, utilizing speech recognition, synthesis, and natural language processing.

Tips

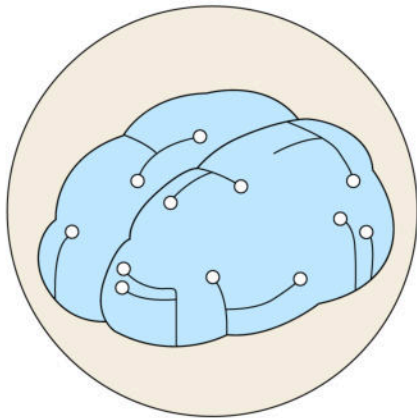
- ✓ **Enhanced Operational Convenience:** Allows craftsmen to keep their hands busy while controlling tools or equipment through voice commands, thereby improving work efficiency and safety.
- ✗ **Speech Recognition Challenges:** Background noise, dialects, and accent differences may cause misinterpretation of commands, which can lead to errors or accidents in precise and complex traditional craftsmanship.

Case



The designer created a visual system based on Guanzhong's language, incorporating Qinqiang, folk songs, and dialects. Centered on "sound," it aims to immerse the audience in the richness of Guanzhong's local culture and artistic expression.

T!



MACHINE LEARNING

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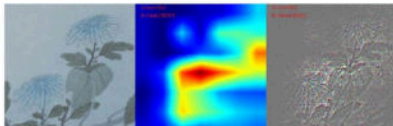
Machine Learning

Machine learning is a key branch of artificial intelligence that enables computer systems to learn from data without explicit programming. In machine learning, algorithms analyze large amounts of data to identify patterns and trends, then use this knowledge to predict future events or make decisions.

Tips

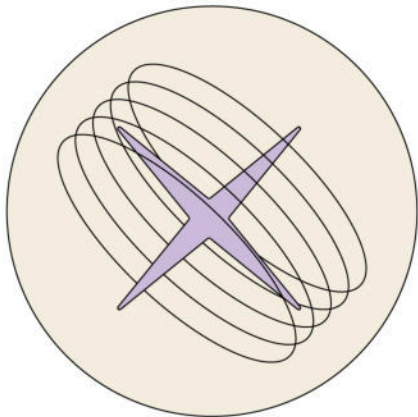
- ✓ **Pattern Recognition and Prediction:** Analyze historical data and apply techniques to identify trends in heritage production, thereby boosting product innovation.
- ✓ **Personalized Product Design:** Customize heritage products based on consumer preferences and behavior data to enhance market competitiveness and customer satisfaction.
- ✗ **Data Collection and Processing Challenges:** Effective models rely on high-quality training data, which can be scarce for rare or ancient heritage items.

Case



This literati painting sentiment dataset utilizes machine learning for sentiment analysis, summarizing the relationship between painting content and the emotions expressed by painters to help the public better understand Chinese culture.

T



HOLOGRAPHY

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Holography

Holography is a technology that presents images or videos in three-dimensional space. By using the principles of light interference and diffraction, holography can transform two-dimensional images into three-dimensional ones, achieving a glasses-free 3D display effect.

Tips

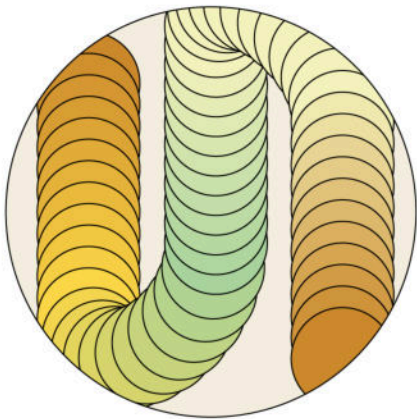
- ✓ **Enhance Interactivity and Immersion:** Holographic systems enable direct interaction with heritage, increasing display engagement and educational value.
- ✗ **Dependency on Sub-Devices:** Relies on electronic devices and specific environmental settings, potentially limiting display locations and conditions.

Case



This performance art employs holographic projection technology in Nuo opera and dance. It allows Nuo culture to overcome limitations related to time, space, scale, and actors, visually presenting its rich imagination through virtual images.

T



BLOCKCHAIN

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Blockchain

Blockchain is a distributed database technology that uses blocks as basic units, storing and managing data through a chain-linking method. Each block contains a number of transaction records and includes the hash value of the previous block, creating an immutable chain structure.

Tips

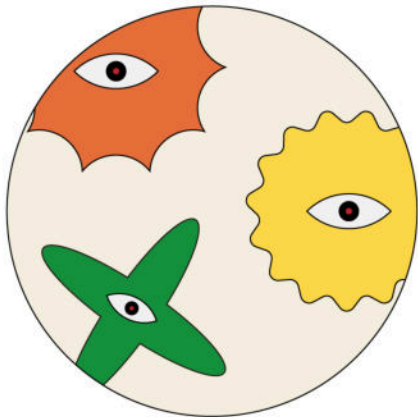
- ✓ **Ensure Transparency and Traceability:** Provide full transparency of heritage origins and production, helping verify authenticity and quality.
- ✓ **Protect Intellectual Property:** Effectively record and safeguard heritage innovations and designs to prevent infringement and imitation.
- ✗ **Regulatory and Legal Uncertainty:** The global legal and regulatory environment for blockchain technology remains unclear, posing potential legal and compliance risks.

Case



This series of digital intangible cultural heritage collections uses blockchain technology. Each artwork has a unique digital certificate, ensuring copyright protection and enabling secure distribution, purchase, collection, and use, thereby expanding cultural consumption channels.

T!



VIRTUAL REALITY

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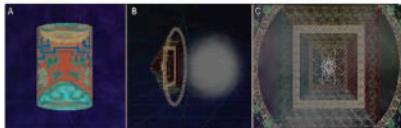
Virtual Reality

Virtual reality (VR) technology is an innovative approach that creates a simulated environment through computer-generated imagery, immersing users as if they are outside the real world. Users can experience virtual environments through VR headsets or immersive displays.

Tips

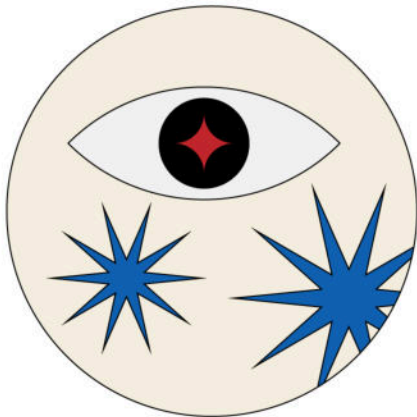
- ✓ **Real-Time Interaction and Modification:** Allows instant interaction and design changes in virtual environments without physical prototypes, accelerating innovation.
- ✓ **Enhanced Interactivity and Immersion:** Helps users better understand the production process of intangible cultural heritage, improving learning efficiency and operational experience.
- ✗ **User Experience Limitations:** Technology cannot fully replicate tactile and olfactory experiences, affecting the completeness and authenticity of user experience.

Case



This Tibetan Buddhist mandala installation was created using virtual reality technology. The installation simulates the meditation experience, giving users a feeling of clearing away mental fog in a meditative state, thereby enriching the tourists' experience.

T!



AUGMENTED REALITY

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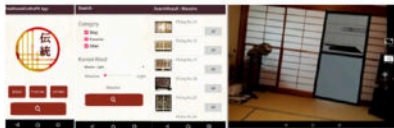
Augmented Reality

Augmented Reality (AR) technology is an innovative method that overlays digital information onto the real world. Users can view their surroundings through devices such as smartphones or AR glasses, with virtual images, videos, or data superimposed onto their view.

Tips

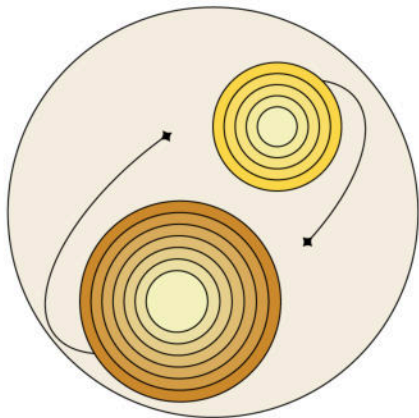
- ✓ **Interactivity and Immersion:** Enhances user experience and understanding of heritage production, improving learning and skill levels.
- ✓ **Real-Time Interaction and Modification:** Enables instant design changes within virtual environments, accelerating innovation without the need for physical prototypes.
- ✗ **User Experience Limitations:** The technology cannot fully replicate the realism and intricate details of heritage items, which may impact the authenticity of the user experience.

Case



This Japanese craft display system uses augmented reality (AR) to showcase traditional items in 3D within real space, allowing users to intuitively place objects and experience their charm.

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INTERNET OF THINGS

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Internet of Things

Internet of Things (IoT) technology is an innovative approach that connects various devices and objects via the Internet to enable real-time data exchange and information sharing. By embedding sensors, communication devices, and data processing capabilities, it allows objects to sense their environment, collect data, and interact.

Tips

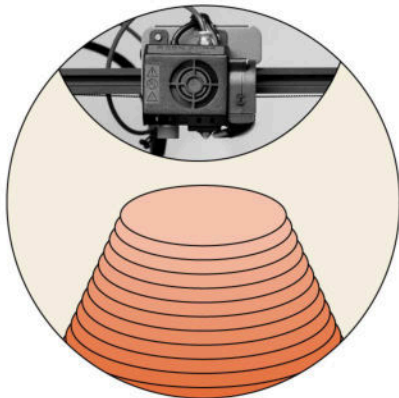
- ✓ **Support flexible customization:** Enables single-piece and small-batch production to meet the market demand for personalized and customized products.
- ✓ **Cost savings:** Real-time monitoring and optimization of production processes, precise control of raw material usage and energy consumption, and ensures product consistency.
- ✗ **Data security and privacy issues:** Data collection and analysis may involve sensitive information, necessitating robust data security measures and compliance to prevent data leakage or abuse.

Case



Silk Worm Box is a smart breeding product that uses IoT technology to simplify silkworm cultivation. Users can monitor and adjust environmental factors like humidity, temperature, and light via a mobile app, addressing the challenge of silkworms' high survival requirements.

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3D PRINTING

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3D Printing

3D printing technology, also known as additive manufacturing, is a method that converts digital models directly into physical objects. By Building objects layer by layer, it's also referred to as layer-by-layer manufacturing. Common types include FDM, MJF, SLS, SLM, DLP, and LCD.

Tips

- ✓ **Rapid Prototyping:** Iterate and innovate quickly to create unprecedented shapes without the limitations of traditional manufacturing processes.
- ✓ **Reduce Material Waste:** Compared to traditional subtractive manufacturing, additive manufacturing generates minimal material waste and is more environmentally friendly.
- ✗ **Material Strength and Durability:** Printed products may not achieve the same mechanical properties as those produced by traditional methods, especially in applications subject to high loads.

Case



Lithobox is a software system, physics kit, and workflow for creating luminescent 3D printed Lithuania. It combines 3D printing technology with traditional porcelain-making techniques, allowing users with no 3D modeling experience to design, print, and assemble luminescent Lithuania.

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Write down your thoughts

